

MOHAMED MEHDI MECHKAK

AI Systems & Design Engineer | Workflow Automation & UI/UX Architecture

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EXECUTIVE PROFILE & SPECIALIZATION

Multi-disciplinary AI Systems & Design Engineer specializing in **autonomous n8n / Make workflow automation**, **production LLM prompt guardrails**, **Apple-standard fluid UI/UX design systems**, and **deterministic data architectures**. Track record of deploying end-to-end client systems that eliminate hundreds of manual operational hours, accelerate user acquisition, and guarantee zero-hallucination execution.

TECHNICAL & DESIGN CAPABILITIES

AI & Automation:	n8n, Make.com, Zapier, Airtable API, Webhooks, LLM System Prompts, Structured Outputs, Anti-Hallucination Guardrails
UI/UX & Design:	Figma to Code, Apple Motion Physics, Tokenized Design Systems, Responsive Layouts, Fluid Sheets, Typography Hierarchy
Frontend & Web:	TypeScript, Next.js 14 (App Router), React, Tailwind CSS, Framer Motion, React Native / Expo, REST & Webhook APIs
Data & Cloud:	Python (FastAPI, Pandas, FFmpeg, Scrapy), Apache Airflow, PostgreSQL, pgvector RAG, Docker, Cloudflare Workers

FLAGSHIP PRODUCTION CASE STUDIES & CLIENT SYSTEMS

StudioPulse & Creative Agency Design System

2026 // Production UI/UX System

Lead Design Engineer & Frontend Architect | [Figma](#) · [TypeScript](#) · [Next.js 14](#) · [Tailwind CSS](#) · [Framer Motion](#) · [Apple Motion Tokens](#)

- Tokenized Figma-to-code design system and interactive masterclass booking engine with Apple fluid motion physics for performing arts studios.
- **Apple Fluid Motion Physics:** Built 1:1 touch-tracked bottom sheet with exponential deceleration velocity projection (0ms input latency).
 - **Dynamic Social Asset Engine:** Programmatic 1080x1920 Instagram Story generator producing on-brand promo graphics in one click.
 - **Impact:** Generated a **3.4x conversion uplift** over traditional static forms; eliminated manual DM booking inquiries.

CreatorFlow & Programmatic Growth Engine

2026 // Content & Video Automation

Growth Automation Architect | [Python](#) · [Next.js 14](#) · [n8n](#) · [FFmpeg](#) · [OpenAI API](#) · [CapCut Pro API](#) · [Webhooks](#)

- Automated short-form video production pipeline, viral hook retention evaluator, and multi-platform publishing dispatcher for creators.
- **Algorithmic Hook Scoring:** Trained 3-second retention blueprint model scoring hook drop-off probability (94% accuracy).
 - **Multi-Channel Dispatcher:** Automated batch formatting and publishing across TikTok, Instagram Reels, and YouTube Shorts.
 - **Impact:** **Saved 20+ hours/week** in video editing overhead; scaled publishing frequency by 3.2x with 100% pipeline automation.

Hotel Concierge AI (ConciergeFlow)

2026 // Hospitality Automation

AI Systems Engineer | [n8n](#) · [Cloudflare Workers](#) · [Airtable API](#) · [Python Harness](#) · [JavaScript](#) · [HTML5/CSS3](#)

- 24/7 AI guest concierge for Paris boutique hotels with grounded n8n workflows and live Airtable inventory integration.
- **Deterministic Integration:** Connected WhatsApp & luxury web chat to live Airtable room availability and guest preferences.
 - **Sub-2s Response:** Cloudflare Worker edge webhook proxy delivering verified multilingual answers with 0% unsourced claims.

PolyLingual AI Support Platform

2026 // Enterprise Support AI

RAG & AI Systems Engineer | [FastAPI](#) · [pgvector](#) · [PostgreSQL](#) · [NVIDIA NIM \(Llama 3.1\)](#) · [Docker](#) · [Pytest](#)

- Dual hybrid lexical + semantic retrieval RAG with confidence-gated fallback routing for customer operations.
- **Confidence Gating:** 78% certainty threshold escalating ambiguous tickets to human staff with pre-drafted context notes.
 - **Operational Impact:** Achieved 88% auto-resolution rate with zero hallucinated policies across 5 languages.

Nexus Control Room (AI Internal Tool Builder)

2025 // Workflow Tooling

Full-Stack Engineer & AI Architect | [TypeScript](#) · [Next.js 14](#) · [Prisma 7](#) · [PostgreSQL](#) · [Tailwind CSS](#) · [Jest](#)

- Natural language to visual node graph generator with server-enforced Prisma PostgreSQL approval state machines.
- **State Machine Security:** Enforced atomic approval transitions (PENDING, APPROVED, RUNNING) blocking unauthorized runs.

EDUCATION & ENGINEERING FOUNDATION

Master of Engineering (Process & Chemical Engineering)

Paris, France

Rigorous engineering discipline: mathematical modeling, system dynamics, thermodynamics, and error propagation analysis.